Time of Exercise Specifies the Impact on Muscle Metabolic Pathways and Systemic Energy Homeostasis

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Article

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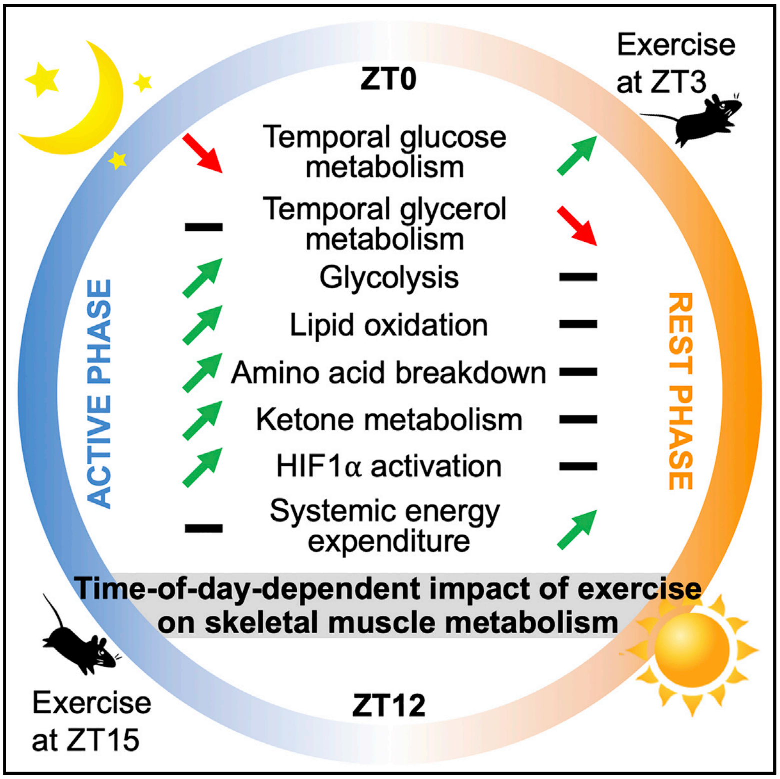
Graphical Abstract



Highlights

- Distinct response of metabolic cycles in skeletal muscle to time-of-day exercise
- Early active phase exercise exerts a robust metabolic response in skeletal muscle
- The metabolic response includes glycolysis, lipid oxidation, and BCAA breakdown
- Time of exercise specifies the activation of HIF1α and systemic energy expenditure

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In Brief

Sato and colleagues investigate time-dependent impact of exercise on skeletal muscle, revealing altered daily metabolic cycles after exercise specific to the time of day. Exercise at the early active phase robustly activates the HIF1α pathway, followed by glycolytic activation, usage of alternative fuels, and adaptation of systemic energy expenditure.

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Time of Exercise Specifies the Impact on Muscle Metabolic Pathways and Systemic Energy Homeostasis

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SUMMARY

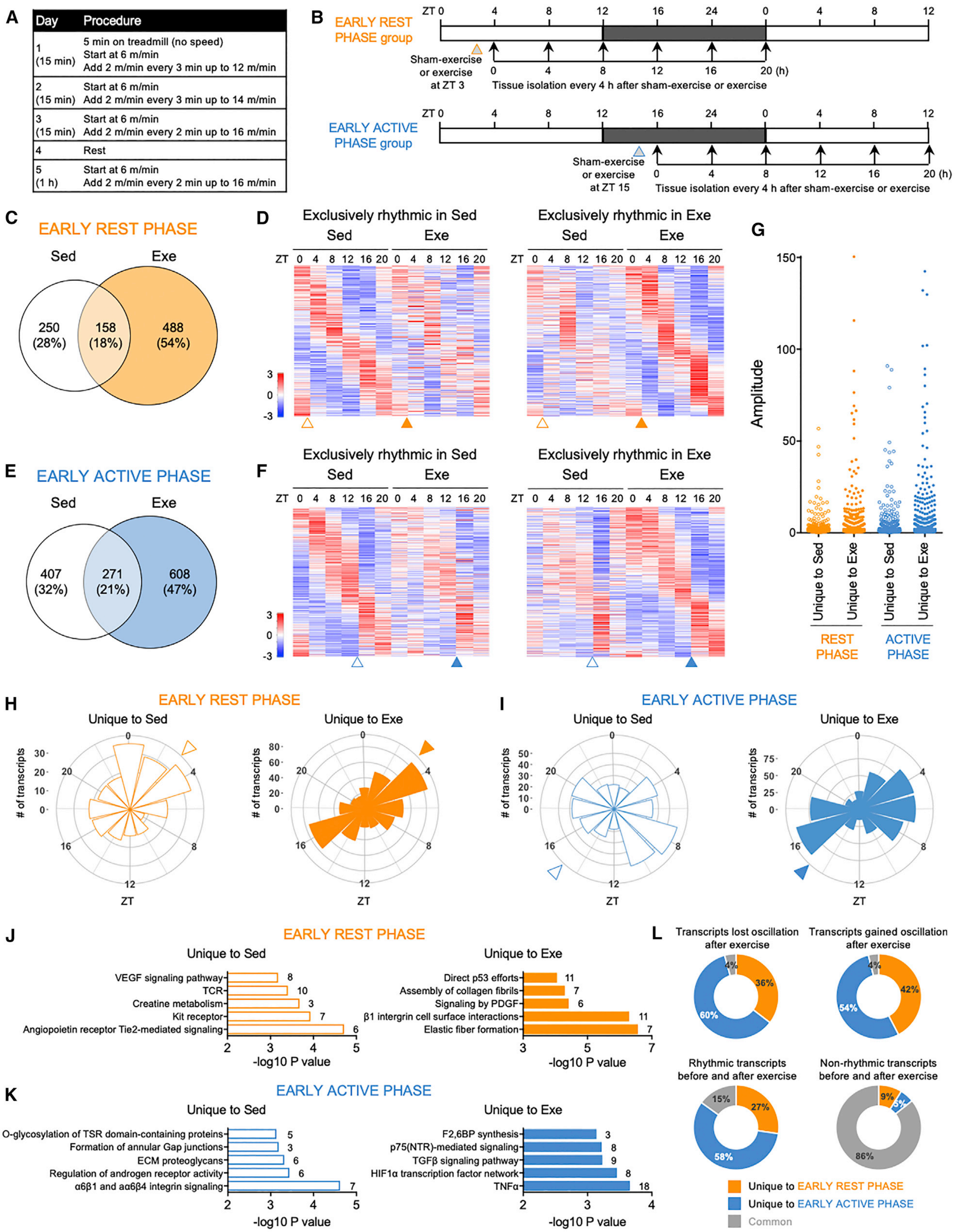
While the timing of food intake is important, it is unclear whether the effects of exercise on energy metabolism are restricted to unique time windows. As circadian regulation is key to controlling metabolism, understanding the impact of exercise performed at different times of the day is relevant for physiology and homeostasis. Using high-throughput transcriptomic and metabolomic approaches, we identify distinct responses of metabolic oscillations that characterize exercise in either the early rest phase or the early active phase in mice. Notably, glycolytic activation is specific to exercise at the active phase. At the molecular level, HIF1α, a central regulator of glycolysis during hypoxia, is selectively activated in a time-dependent manner upon exercise, resulting in carbohydrate exhaustion, usage of alternative energy sources, and adaptation of systemic energy expenditure. Our findings demonstrate that the time of day is a critical factor to amplify the beneficial impact of exercise on both metabolic pathways within skeletal muscle and systemic energy homeostasis.

INTRODUCTION

Many biological processes are characterized by daily variations that are controlled by the circadian clock (Bass and Lazar, 2016; Top and Young, 2018). The central clock located in the hypothalamic suprachiasmatic nucleus (SCN) and entrained by light as an external *zeitgeber* (time-giver) contributes to synchronize peripheral clocks (Akhtar et al., 2002; Cheng et al., 2002; Hastings et al., 2018; LeGates et al., 2014; Panda et al., 2002). Diverse environmental cues such as feeding also act as robust *zeitgebers* for peripheral clocks in metabolic tissues through mechanisms that appear to work independently of SCN (Asher and Sassone-Corsi, 2015; Damiola et al., 2000; Hatori et al., 2012; Panda, 2016; Vollmers et al., 2009). Importantly, metabolic coordination and communication between clocks is essential for homeostatic balance (Dyar et al., 2018b). These notions parallel findings illustrating the intertwined connection between the circadian clock and metabolism: the circadian clock directly influences cyclic metabolic functions, while small cellular metabolites such as ATP, nicotinamide adenine dinucleotide (NAD^+), acetyl-CoA, S-adenosylmethionine (SAM), and glucose are involved in the maintenance of cellular rhythmicity (Bass and Takahashi, 2010; Green et al., 2008; Katada et al., 2012; Nakahata et al., 2009; Ramsey et al., 2009). Finally, metabolic pathways controlled by the circadian clock are reprogrammed by dietary challenges (Eckel-Mahan et al., 2013; Guan et al., 2018; Kinouchi

Context and Significance

Exercise helps the improvement of metabolic health. While the metabolic benefits from exercise have been extensively demonstrated, the question of when it is appropriate to exercise has remained virtually unexplored. Sato and colleagues have compared the impact of exercise at different times of the day on skeletal muscle metabolism in mice. The authors observed a more robust metabolic impact of exercise in the morning (beginning of active phase) than at night (beginning of rest phase), resulting in a higher utilization of carbohydrates and ketone bodies, together with the degradation of lipids and amino acids. This study suggests that the timing of exercise during the day may prove to be a valuable therapy for patients with metabolic disorders.



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et al., 2018; Kohsaka et al., 2007; Sato et al., 2017; Tognini et al., 2017).

Physical exercise is beneficial for skeletal muscle metabolism and systemic energy homeostasis (Baskin et al., 2015; Vega et al., 2017). Skeletal muscle adaptation to exercise implicates a variety of molecular pathways controlling muscle contraction coupled to ATP biosynthesis as well as energy utilization by the activation of mechanosensors and metabolic sensors. These include mitogen-activated protein kinases (MAPKs), AMP-activated protein kinase (AMPK), and Sirtuins (SIRTs), as well as the oxygen sensors prolyl hydroxylases (PHDs) (Camera et al., 2016; Egan and Zierath, 2013; Fan and Evans, 2017; Fan et al., 2017; Kjøbsted et al., 2018). The sensors transduce exercise-induced stimuli to downstream transcription factors that activate target genes involved in glucose metabolism, lipid metabolism, and mitochondrial biogenesis. Overall, exercise is widely recognized as a beneficial therapeutic strategy for human health through the remodeling of specific metabolic signaling pathways (Hawley et al., 2018; Overmyer et al., 2015). Importantly, in addition to the quantity, duration, and type of exercise, the time of exercise is suggested to be one of the variables for physiological outputs from exercise (Gabriel and Zierath, 2019; Sasaki et al., 2014; Schroeder et al., 2012). Yet, the question of time-specific impact of exercise on metabolic pathways remains virtually unexplored.

The circadian clock within skeletal muscle plays a critical role in the regulation of muscle metabolism (Dyar et al., 2014, 2018a; Jordan et al., 2017; Perrin et al., 2018). Moreover, the function of the circadian clock is tightly linked to exercise-stimulated metabolic sensors, including AMPK and SIRTs (Lamia et al., 2009; Masri and Sassone-Corsi, 2014; Nakahata et al., 2008), pointing to the potential role played by exercise in the temporal regulation of metabolic pathways. Importantly, hypoxia as well as the accumulation of tricarboxylic acid (TCA) cycle intermediates suppresses the activity of PHDs, leading to hypoxia-inducible factor 1α (HIF1α) stabilization and enhanced transcription of HIF1α-driven genes, including genes involved in glycolysis (Favier et al., 2015; Ke and Costa, 2006; Koivunen et al., 2007; Pan et al., 2007; Schödel et al., 2011). This contributes to metabolic shift to glycolysis rather than oxidative phosphorylation under hypoxic conditions during exercise (Lindholm and Rundqvist, 2016). In addition to the metabolic role of HIF1α, HIF1α also coordinates circadian clock activity (Adamovich et al., 2017; Peek et al., 2017; Wu et al., 2017). These findings underscore the possibility that exercise might affect temporal control of metabolism through the activation of metabolic sensors as well as of the hypoxia-responsive HIF1α transcriptional pathway.

How the timing of exercise may influence metabolic pathways to elicit the most beneficial effects remains an outstanding question. This question would be ultimately applicable to human physiology and thereby might have important therapeutic implications. By performing high-throughput transcriptomics and metabolomics along the daily cycle, we compared specific signatures in mouse skeletal muscle after acute exercise performed at the early rest phase versus early active phase. We reveal that exercise triggers metabolic responses in a time-dependent manner, demonstrating that the time of exercise is a critical modulator for skeletal muscle metabolism and systemic energy homeostasis.

RESULTS

Daily Transcriptional Programs upon Exercise Differ at Early Rest and Early Active Phases

To determine whether exercise has a time-of-day-specific impact on circadian regulation of metabolic pathways in skeletal muscle, mice were divided into 4 groups including sedentary control (Sed) and exercise (Exe) at the early rest phase (ZT3), and Sed and Exe mice at the early active phase (ZT15). Mice were sacrificed at 0, 4, 8, 12, 16, and 20 h after a single 1-h bout of acute exercise or sham exercise on a stationary treadmill (Figures 1A and 1B). At the early rest phase, 250 genes (28%) are rhythmic only in Sed mice, whereas 488 genes (54%) gain *de novo* oscillation after exercise (Figures 1C, 1D, and S1A; Data S1). While the overlap of the cyclic transcripts between Sed and Exe mice at the active phase is only 21% of total cyclic transcripts, 407 (32%) and 608 (47%) genes exhibit daily oscillation unique to Sed and Exe mice, respectively (Figures 1E, 1F, and S1B; Data S1). Amplitude analysis shows that exercise at either the rest or active phase positively impacts the amplitude of

Figure 1. Response of the Temporal Transcriptome in Skeletal Muscle to Exercise at Early Rest Phase versus Early Active Phase

(A) Exercise protocol applied in this study. The mice were subjected to a single bout of acute treadmill running following a 4-day acclimatization protocol.

(B) Experimental design for the acute exercise at the early rest phase (ZT3) and the early active phase (ZT15) and tissue isolation every 4 h after sham exercise or exercise.

(C) Venn diagram displaying the number (top) and ratio (bottom) of oscillating genes in skeletal muscle isolated from sedentary (Sed) mice and exercised (Exe) mice at the early rest phase (ZT3).

(D) Heatmaps displaying rhythmic transcripts unique to Sed (left panel) and Exe (right panel).

(E) Venn diagram displaying cyclic genes from Sed and Exe mice at the early active phase (ZT15).

(F) Heatmaps displaying rhythmic genes found exclusively in Sed (left panel) and Exe (right panel).

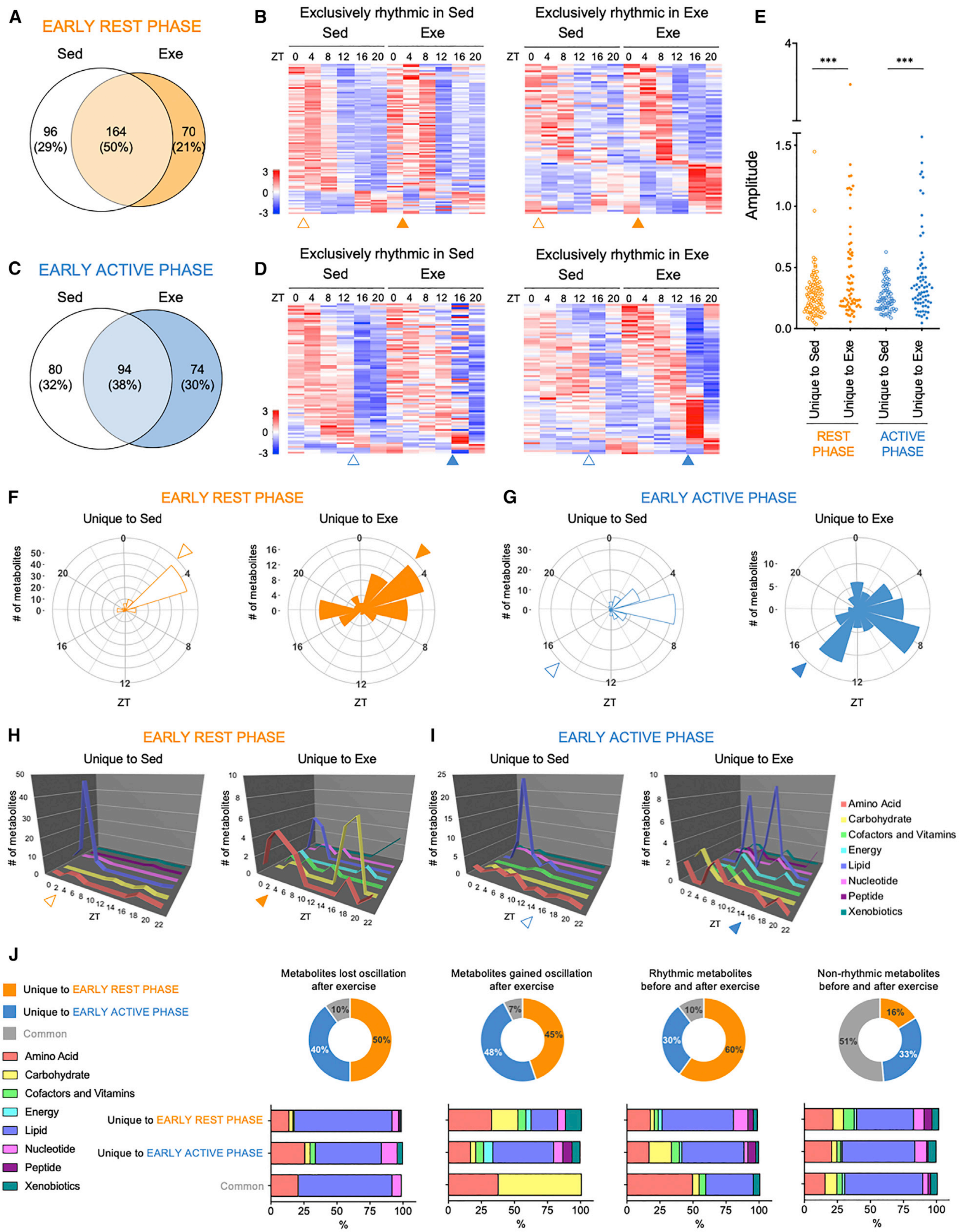
(G) Amplitude distribution of oscillating genes in Sed and Exe mice at rest and active phase, respectively.

(H and I) Radar plots representing the phase lag of rhythmic genes unique to Sed (left panel) and Exe (right panel) mice at rest phase (H) and active phase (I).

(J and K) GO analysis of cyclic genes unique to Sed (left) and Exe (right) at rest phase (J) and active phase (K). TCR, T-cell receptor; PDGF, platelet-derived growth factor; ECM, extracellular matrix; TSR, thrombospondin type-1 repeat; TNF-α, tumor necrosis factor α; HIF1α, hypoxia-inducible factor 1α; TGF-β, transforming growth factor β; p75(NTR), p75 neurotrophin receptor; and F2,6BP, fructose 2,6-bisphosphate.

(L) Donut plots exhibiting the overlap of transcripts that oscillate only in Sed (transcripts lost oscillation), only in Exe (transcripts gained oscillation), both in Sed and Exe (rhythmic transcripts), and neither in Sed and Exe (non-rhythmic transcripts) between rest and active phase.

The transcriptomic data contain 5–6 individual gastrocnemius muscles per time point. The data were placed from ZT0 to ZT20 in the heatmaps and radar plots. Triangles indicate the timing of sham exercise or exercise. Nonparametric test Bio_Cycle was applied for the detection of the significance of daily oscillation using p value cutoff of 0.05.



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oscillating muscle transcripts (Figures 1G, S1C, and S1D). In addition to altering the amplitude of cycling transcripts, exercise also induces robust redistribution of the phase of oscillating genes in skeletal muscle (Figures 1H, 1I, S1E, and S1F).

To reveal the daily biological signature of exercise, we carried out Gene Ontology (GO) analysis of muscle transcripts oscillating exclusively in Sed versus Exe mice at each phase (Figures 1J and 1K). Notably, in the early active phase group, the inflammatory signaling pathway as well as the HIF1α regulatory pathway is specifically enriched in Exe mice (Figure 1K). Common cycling transcripts in skeletal muscle from Sed and Exe mice exhibit much higher enrichment of genes engaged in clock control, regardless of the timing of exercise or sham-exercise treatment (Figures S1G and S1H). Comprehensive genome-wide mapping of transcriptional regulatory elements (Daily et al., 2011; Xie et al., 2009) revealed that circadian transcription factors CLOCK-BMAL1 and RORα as well as myogenic transcriptional factors MyoD and Myogenin are highly enriched in oscillating genes common to Sed and Exe mice in both phase groups (Figures S1I and S1J; Table S1). Remarkably, HIF1 and HIF2α are present as unique factors in rest phase Sed mice, whereas genes with HIF1 target sites are enriched in active phase Exe mice (Figures S1I and S1J). These observations correlate with the GO analysis of rhythmic genes, indicating a time-of-day-dependent response of hypoxia-induced transcriptional regulation to exercise.

To reveal how the timing of exercise differently influences the daily transcriptome, we compared the proportion of transcripts that lost oscillation, gained oscillation, remained oscillatory, or remained non-rhythmic after exercise in the early rest phase versus early active phase (Figure 1L). Remarkably, only 4% of transcripts that lost or gained rhythmicity in response to exercise were common to both phases. Thus, these results clearly demonstrate the time-dependent effect of exercise on the daily transcriptome of skeletal muscle.

Most of core circadian clock components remain oscillatory even after exercise, both in the early rest phase and in the early active phase (Figures S1K and S1L). Interestingly, rhythmic expression of *Per1* is virtually absent at the early rest phase, regardless of the presence or absence of an exercise bout (Figure S1K), while *Cry1* and *Cry2* exhibit significant oscillation only in Exe mice from the early active phase group (Figure S1L). Additional comparison of daily transcriptome in Sed animals between the early rest phase group and the early active phase group reveals that only 15% of cycling genes are common to the rest and active phase groups (Figure S2A) but most of the core clock genes oscillate in both groups (Figures S2B and S2C). Importantly, mice exposed to a stress stimulus at different times of day exhibit a distinct response to the circadian clock system in peripheral tissues (Tahara et al., 2015), supporting the notion of distinct daily transcriptional outputs in the rest phase versus active phase. Indeed, we show that there is only a marginal overlap of rhythmic transcripts in Exe mice between the early rest phase and the early active phase (Figure S2D), whereas core clock genes are common to both phases (Figures S2E and S2F). These results also highlight the importance of using Sed animals for each phase and underscore that the comparison of data from temporal high-throughput analyses may be appropriate only within the same phase group.

Temporal Metabolomics Reveals Time-Dependent Impact of Exercise on Metabolic Cycles

The time-specific effect of exercise on metabolic pathways illustrated by the transcriptomics analysis prompted us to extend our study to high-throughput metabolomics. Mass spectrometry (MS) of muscle samples detected 672 biochemicals, with 580 compounds of known identity and 92 compounds of unknown structural identity. Principal component analysis (PCA) identified the separation of samples based on the time of collection between Sed and Exe mice (ZT4 and ZT8, 0 and 4 h after exercise or sham-exercise treatment in the early rest phase group, and ZT16 and ZT20, 0 and 4 h after exercise or sham-exercise treatment in the early active phase group) (Figure S3A). In the early rest phase group, 330 metabolites oscillate overall, of which 96 (29%) and 70 (21%) metabolites are unique to Sed and Exe mice, respectively, whereas 164 (50%) metabolites oscillate irrespective of exercise (Figures 2A, 2B, and S3B; Data S2). Of 248 rhythmic metabolites in the early active phase group, 80 (32%) and 74 (30%) metabolites are found exclusively rhythmic in Sed and Exe mice, respectively, whereas 94 (38%) metabolites oscillate both in Sed and Exe mice (Figures 2C, 2D, and S3C; Data S2). Strikingly, amplitude analysis reveals that exercise significantly fortifies the amplitude of cycling metabolites independently of whether exercise is performed during the rest or active phase (Figures 2E, S3D, and S3E), suggesting that exercise could amplify the daily skeletal muscle metabolome. The temporal metabolome also reveals the distinct signature of the

Figure 2. Impact of Exercise at Rest versus Active Phase on the Temporal Metabolome in Skeletal Muscle

(A) Venn diagram displays the number (top) and ratio (bottom) of oscillating metabolites in skeletal muscle isolated from sedentary (Sed) mice and exercised (Exe) mice at the early rest phase (ZT3).

(B) Heatmaps display rhythmic metabolites unique to Sed (left panel) and Exe (right panel).

(C) Venn diagram displays cycling metabolites from the Sed and Exe groups at the early active phase (ZT15).

(D) Heatmaps display cycling metabolites found exclusively in Sed (left panel) and Exe (right panel).

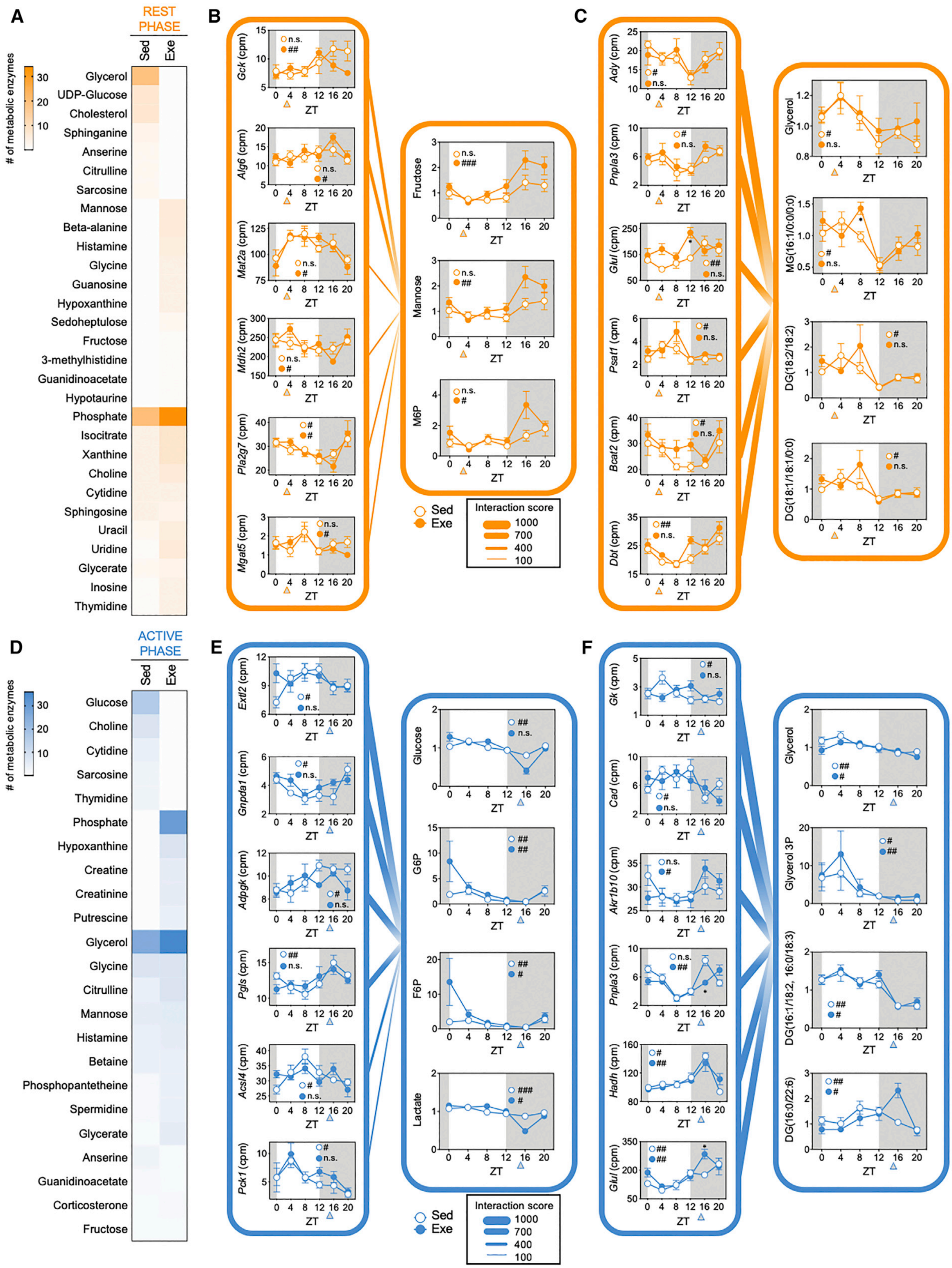
(E) Amplitude distribution of rhythmic metabolites in Sed and Exe mice at rest and active phase, respectively. ***$p < 0.001$ analyzed by Student's t test.

(F and G) Radar plots represent the phase lag of rhythmic metabolites unique to Sed (left panel) and Exe (right panel) mice at rest phase (F) and active phase (G).

(H and I) Temporal metabolites landscape of rhythmic metabolites exclusively in Sed (left) and Exe (right) at rest phase (H) and active phase (I).

(J) Donut plots exhibit the overlap of metabolites that oscillate only in Sed (metabolites lost oscillation), only in Exe (metabolites gained oscillation), both in Sed and Exe (rhythmic metabolites), and neither in Sed and Exe (nonrhythmic metabolites) between rest and active phase. Stacked bar charts display the phase-specific composition of biological classifications of metabolites that oscillate only in Sed, only in Exe, both in Sed and Exe, and neither in Sed and Exe.

The metabolomic data contain 6 individual gastrocnemius muscles per time point. The data were placed from ZT0 to ZT20 in the heatmaps, radar plots, and temporal metabolites landscapes. Triangles indicate the timing of sham exercise or exercise. Nonparametric test Bio_Cycle was applied for the detection of the significance of daily oscillation using p value cutoff of 0.05.



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phase of oscillatory metabolites after exercise at the early rest phase versus the early active phase (Figures 2F, 2G, S3F, and S3G). These observations underpin not only the massive impact of exercise on phase redistribution of the temporal metabolome but also the interplay between the daily skeletal muscle transcriptome and metabolome after exercise as unique to the early rest versus active phase.

Next, we determined temporal profiles of metabolites related to biological classes. Skeletal muscle isolated from Sed animals from both rest and active phase groups has an intensive peak of metabolites involved in lipid metabolism at the early light phase (Figures 2H and 2I). While rest phase exercise does not alter the timing of the peak of lipid metabolites, exercise at the early active phase induces an immediate *de novo* peak of lipid metabolites (Figures 2H and 2I). Remarkably, exercise at the early rest phase also induces a peak of amino acids immediately after exercise, as well as carbohydrates peaking at ZT16, 12 h after exercise cessation (Figure 2H). In comparison, exercise at the early active phase does not significantly affect amino acids but rather induces a peak of carbohydrates at the beginning of the light phase (Figure 2I), suggesting that feeding activity after exercise exhibits time-of-day dependency. Rhythmic metabolites common to Sed and Exe mice in each rest and active phase group mostly relate to lipid metabolism and, to a lesser extent, amino acid metabolism (Figures S3H and S3I). These results indicate that energy utilization in skeletal muscle during or after exercise is different depending on the time of day.

To further demonstrate the time-of-day-specific impact of exercise on the temporal metabolome, we determined the overlap of rhythmic and non-rhythmic metabolites before and after exercise at the early rest phase versus early active phase (Figure 2J). Similar to the daily transcriptome, there is a manifest time dependence of the effect on metabolite oscillations. Only 10% and 7% of metabolites lose or gain daily oscillations in a timing-of-exercise-independent manner (Figure 2J). Interestingly, most of these common exercise-sensitive metabolites that lost rhythmicity after exercise are lipids, whereas metabolites that gained oscillation are mostly carbohydrates (Figure 2J). In stark contrast, metabolites that lost (50%) or gained oscillation (45%) after exercise are unique to the rest phase, while 40% lost and 48% gained oscillation in the active phase (Figure 2J). Metabolites that lost oscillation after exercise relate mostly to lipid and amino acid metabolism, for both the early rest and early active phase exercise (Figure 2J). Thus, the timing determines how exercise modulates rhythmic metabolism in skeletal muscle.

Linking Cycling Metabolic Enzymes and Metabolites Is Unique to Time of Exercise

We next performed a combined analysis of the daily transcriptome and metabolome in skeletal muscle to reveal the coherence between metabolic enzymes identified by the temporal transcriptome and metabolites identified by the temporal metabolome (Table S2). Each enzyme-metabolite pair was assigned a score indicative of likelihood of interaction. In the early rest phase group, enzymes involved in glycerol metabolism are highly enriched in Sed mice, whereas enzymes involved in mannose metabolism are unique to Exe mice (Figure 3A). Indeed, exercise at the early rest phase enhances the daily oscillation of genes related to mannose metabolism as well as the daily oscillation of mannose, mannose-6-phosphate (M6P), and fructose (Figure 3B). Strikingly, exercise at the early rest phase severely dampens daily oscillations of genes and metabolites involved in glycerol metabolism (Figure 3C). The GO analysis of the metabolic enzymes showing significant interaction between the daily transcriptome and metabolome in early rest phase group reveals higher enrichment of lipid metabolic processes in Sed mice but glycolytic processes in Exe mice (Figure S4A).

In the early active phase group, the enrichment of metabolic enzymes related to glucose metabolism is higher in Sed mice (Figure 3D). After exercise at the early active phase, there is a loss of significant oscillation of genes encoding glycolytic enzymes as well as glycolytic intermediate metabolites (Figure 3E), uncovering a distinct response of the cyclic carbohydrate metabolism to exercise at the early rest phase versus the early active phase (Figures 3B and 3E). In contrast to the early rest phase exercise, strong interactions remain between the rhythmic transcripts and metabolites related to glycerol upon exercise at the early active phase (Figure 3F), suggesting that cyclic glycerol metabolism is less responsive to exercise at the early active phase compared to the early rest phase. GO analysis of coherent enzymes between the daily transcriptome and metabolome in the early active phase group confirms the glycolytic process as a daily metabolic pathway unique to Sed mice and lipid metabolic processes as daily metabolic pathways common to both Sed and Exe mice in the early active phase group (Figure S4B). Thus, timing of exercise is a critical factor to specify cyclic metabolic pathways, showing robust

Figure 3. Interlocked Signatures between the Daily Transcriptome to Metabolome in Skeletal Muscle

(A) Heatmap displays the number of interacting metabolic enzymes identified by the temporal transcriptome with each class of metabolites identified by the temporal metabolome in Sed versus Exe mice at the early rest phase.

(B) Typical examples representing the interaction between daily oscillation of transcripts and metabolites involved in glucose metabolism in Exe mice at the early rest phase.

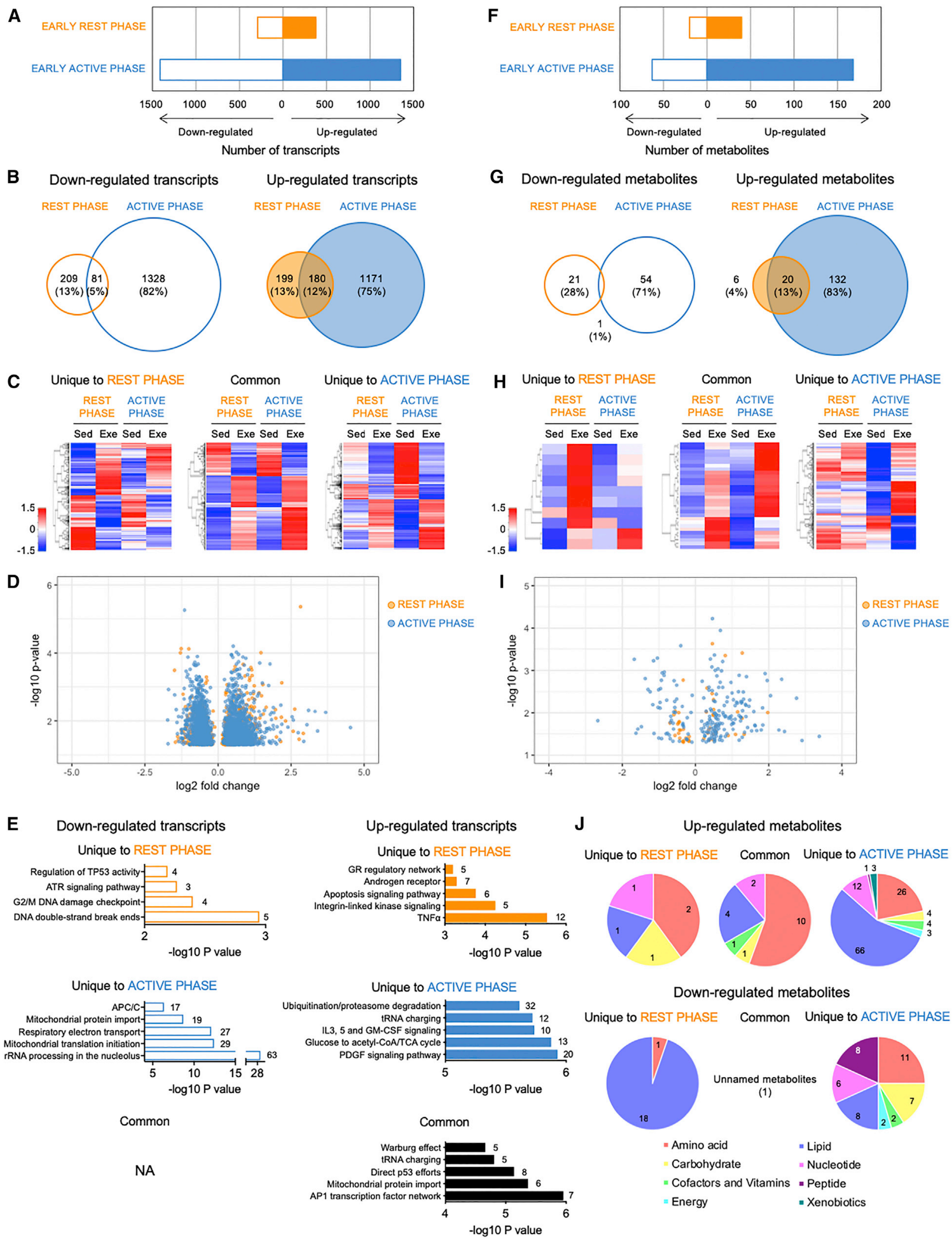
(C) Coherence between daily oscillation of transcripts and metabolites involved in glycerol metabolism in Sed mice at the early rest phase.

(D) Heatmap displays the number of interacting metabolic enzymes identified by the temporal transcriptome with each class of metabolites identified by the temporal metabolome in Sed versus Exe mice at the early active phase.

(E) Correlation between daily oscillation of transcripts and metabolites involved in glucose metabolism in Sed mice at the early active phase group.

(F) Interaction between daily oscillation of transcripts and metabolites involved in glycerol metabolism both in Sed and Exe mice at the early active phase.

Interacting scores between oscillating transcripts and metabolites were calculated via the CircadiOmics web portal (Ceglia et al., 2018; Patel et al., 2012). Data are represented as mean ± SEM (n = 5–6/time point). The transcripts and metabolites data were placed from ZT0 to ZT20. Triangles indicate the timing of sham exercise or exercise. The significance of the daily oscillation of genes and metabolites was detected by Bio_Cycle, based on a p value cutoff of 0.05. #$p < 0.05$, ##$p < 0.01$, and ###$p < 0.001$.



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coherence between the temporal transcriptome and metabolome in skeletal muscle.

Time-of-Day-Specific Acute Response to Exercise in Skeletal Muscle

The responses of cellular signaling and subsequent gene transcription to exercise are very rapid but very transient (Egan and Zierath, 2013). It is, therefore, predictable that exercise exerts acute transcriptomic and metabolomic responses. In addition to the temporal analyses to uncover the impact of exercise at the early rest phase versus the early active phase on daily rhythmicity of the transcriptome and metabolome, we next probed time-of-day specificity of acute response of the muscle transcriptome and metabolome to exercise. We first compared the differences in the levels of transcripts between skeletal muscle isolated from mice immediately after sham-exercise treatment or acute exercise. Remarkably, 1,351 (11.8% of total transcripts) and 1,409 genes (12.3% of total) are significantly increased and decreased, respectively, after the early active phase exercise, whereas only 379 (3.3% of total) and 290 genes (2.5% of total) are significantly increased and decreased, respectively, after the early rest phase exercise (Figure 4A; Data S3). There is a trivial overlap of transcripts changed after exercise at the early rest phase versus the early active phase, demonstrating the time-of-day-dependent acute response of gene transcription to exercise (Figures 4B–4D). GO analysis of downregulated and upregulated transcripts revealed that rRNA processing, as well as processes involved in the regulation of mitochondrial function, such as mitochondrial translation, respiratory electron transport, and mitochondrial protein import, are highly enriched within transcripts exclusively downregulated after the early active phase exercise (Figures 4E and S5A), suggesting dampened mitochondrial respiratory function after exercise at the early active phase. Upregulated transcripts only after exercise at the early rest phase exhibit significant enrichment in genes involved in inflammatory response and apoptotic signaling, while the platelet-derived growth factor (PDGF) signaling pathway and the glycolytic pathway are highly enriched within transcripts exclusively upregulated by the active phase exercise (Figures 4E and S5B). Of note, the top 2 biological processes within upregulated genes unique to active phase exercise (PDGF signaling pathway and glycolytic pathway) have been shown to be under the control of HIF1α (Favier et al., 2015; Heber-Katz, 2017; Majmundar et al., 2010). This indicates time-dependent-HIF1α activation and subsequent stimulation of angiogenesis and glycolysis in skeletal muscle after exercise. These observations show distinct transcriptional responses of skeletal muscle to exercise, which are unique to the time of day.

Similar to the observation from the transcriptomic analysis, an analysis of the muscle metabolome identified a larger number of exercise-sensitive metabolites at the active phase than the rest phase. Only 22 (3.3% of total metabolites) and 26 metabolites (3.9% of total) are downregulated and upregulated, respectively, immediately after the early rest phase exercise, whereas 55 (8.2% of total) and 152 metabolites (22.6% of total) are downregulated and upregulated, respectively, by the early active phase exercise (Figure 4F; Data S3). A marginal overlap is observed between metabolites altered after exercise at the early rest phase versus the early active phase (Figures 4G–4I). The biological classification of metabolites shows higher enrichment of lipids and amino acids as well as insignificant enrichment of carbohydrates within metabolites increased only by the active phase exercise (Figure 4J). In contrast, the downregulation of metabolites involved in carbohydrate metabolism after active phase exercise is more remarkable (Figure 4J). Almost all of the downregulated metabolites unique to the rest phase exercise are lipid metabolites (Figure 4J). Parallel to the observations from the transcriptome analysis, the skeletal muscle metabolome exhibits a distinct exercise response that is dependent on the time of day.

Time-of-Exercise-Dependent Activation of the Glycolytic Pathway in Skeletal Muscle

The results obtained with the transcriptome and metabolome analyses clearly demonstrated a robust metabolic impact of exercise at the early active phase rather than the early rest phase. Consistent with the transcriptomic and metabolomic findings, muscle and blood glucose levels are decreased only after exercise at the early active phase (Figures 5A and 5B). Parallel to the changes in glucose levels, a reduction in muscle glycogen content is observed only after exercise at the early active phase (Figure 5C). Interestingly, hepatic glycogen content is significantly decreased after exercise at the early rest phase (Figure 5D).

Figure 4. Distinct Acute Impact of Exercise at Rest versus Active Phase on Muscle Transcriptome and Metabolome

(A) The number of increased and decreased transcripts in skeletal muscle immediately after exercise at early rest phase (ZT3) versus early active phase (ZT15).

(B) Venn diagrams representing the overlap of transcripts that downregulated (left) or upregulated (right) by exercise between rest and active phase.

(C) Heatmaps display the response of skeletal muscle transcripts to exercise unique to rest phase (left), unique to active phase (right), and common to both phases (middle).

(D) Volcano plot exhibits the significance of altered muscle transcripts by exercise at rest (orange dots) and active phase (blue dots).

(E) GO analysis of downregulated (left) and upregulated (right) muscle transcripts by exercise unique to rest phase (top) and active phase (middle), and common to both phases (bottom). Numbers within the charts indicate number of transcripts identified within each biological pathway. ATR, ataxia telangiectasia and Rad3-related protein; GR, glucocorticoid receptor; APC/C, anaphase-promoting complex/cyclosome; and GM-CSF, granulocyte macrophage colony-stimulating factor.

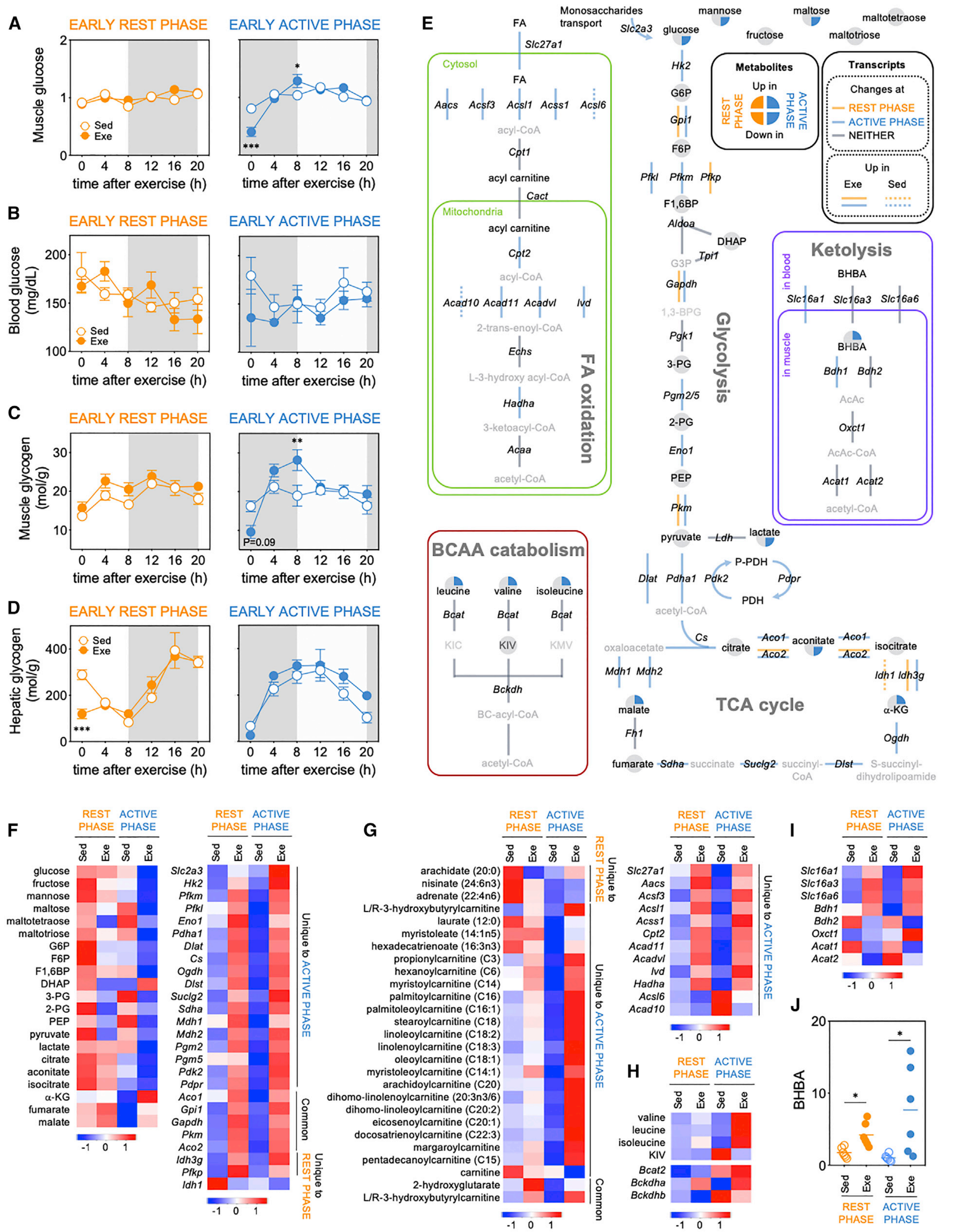
(F) The number of increased and decreased metabolites in skeletal muscle immediately after exercise at the early rest phase (ZT3) versus active phase (ZT15).

(G) Venn diagrams representing the overlap of metabolites that downregulated (left) or upregulated (right) by exercise between rest phase and active phase.

(H) Heatmaps display the response of skeletal muscle metabolites to exercise unique to rest phase (left), unique to active phase (right), and common to both phases (middle).

(I) Volcano plot exhibits the significance of altered muscle metabolites by exercise at rest phase (orange dots) and active phase (blue dots).

(J) Pie charts show the biological classifications of muscle metabolites exclusively upregulated (top) or downregulated (bottom) by exercise at rest phase (left), active phase (right), and both phases (middle). Numbers within the pie charts indicate the number of metabolites identified within each biological classification. The transcriptomic and metabolomic data contain 6 individual gastrocnemius muscles per group.



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Considering the daily variation of hepatic glycogen concentration derived from the feeding-fasting cycle, hepatic glycogen might be utilized during exercise when the hepatic glycogen concentration is high, whereas hepatic glycogen might be sparred during exercise when the hepatic glycogen content is low. These results strongly suggest a time-of-day-dependent impact of exercise on glucose metabolism unique to skeletal muscle.

We next carried out a combined analysis of the transcriptome and metabolome to identify enzyme-encoding transcripts and metabolites that are upregulated or downregulated after exercise at the early active phase compared to the early rest phase (Figure 5E). We found a robust reduction in muscle metabolites related to carbohydrate parallel to an induction of muscle transcripts involved in glycolysis only by exercise at active phase (Figures 5E and 5F). In addition to metabolites involved in glycolysis, intermediates in the first part of the TCA cycle declined after the active phase exercise, whereas the induction of genes encoding TCA cycle enzymes is greater after exercise at the active phase than the rest phase (Figures 5E and 5F). In connection to these observations, the active phase exercise elevates the levels of intermediates in the later part of the TCA cycle, such as α-ketoglutaric acid (α-KG), fumarate, and malate (Figures 5E and 5F), indicating the accumulation of TCA cycle intermediates and mitochondrial respiratory dysfunction during exercise at the early active phase. These results demonstrate that during exercise in the early active phase, skeletal muscle relies on the glycolytic pathway to generate acetyl-CoA for oxidation in the mitochondrial TCA cycle for energy production.

Acetyl-CoA is placed at the intersection of a number of metabolic pathways, including glycolysis, fatty acid oxidation, amino acid degradation, and ketone body metabolism (Shi and Tu, 2015; Sivanand et al., 2018). In addition to carbohydrates, fatty acids, amino acids, and ketone bodies can be utilized as energy sources by skeletal muscle during exercise through the biosynthesis of acetyl-CoA. Strikingly, exercise during the early active phase promotes the levels of muscle transcripts encoding enzymes and transporters involved in fatty acid oxidation (Figures 5E and 5G). More remarkably, the levels of acylcarnitine, fatty acids bound to carnitine for the translocation into mitochondria, are dramatically elevated only after the early active phase exercise (Figure 5G). These results clearly demonstrate a time-of-day-dependent impact of exercise on the activation of fatty acid oxidation in skeletal muscle. Similarly, the levels of branched-chain amino acids and genes involved in amino acid degradation are preferably elevated after exercise at the early active phase (Figures 5E and 5H). Finally, a significant upregulation of genes encoding ketone body transporter (*Slc16a1*) as well as genes involved in ketolysis (*Bdh1*) is observed after exercise at the early active phase (Figure 5I). Conclusively, a massive increase in the levels of ketone body β-hydroxybutyrate (BHBA) in skeletal muscle is observed after exercise at the early active phase (7.5-fold) compared to the early rest phase (2.4-fold) (Figure 5J). Thus, energy utilization in skeletal muscle during exercise is highly dependent on the time of day. Moreover, exercise during the early active phase might require multiple energy sources, including carbohydrates, lipids, amino acids, and ketone bodies in skeletal muscle.

Exercise at the Early Active Phase Activates HIF1α Signaling Pathway

Skeletal muscle is one of the most oxygen-dependent metabolic tissues in the body. Type I and type IIA muscle fibers highly rely on oxidative metabolism to generate ATP required for muscle contraction. Conversely, skeletal muscle can readily adapt to multiple external stressors, including hypoxia (Favier et al., 2015; Lindholm and Rundqvist, 2016). Indeed, the activation of glycolytic enzymes is observed in the skeletal muscle in response to hypoxia (Favier et al., 2015). Importantly, the oxygen-sensitive transcription factor HIF1α directly binds to the promoters of genes involved in the glycolytic pathway (Schödel et al., 2011). Therefore, we next pursued the relationship between time-of-day-dependent activation of HIF1α and glycolysis. Glycolytic HIF1α target genes exhibit higher enrichment of genes upregulated after the early active phase exercise than the early rest phase exercise (Figure 6A). In addition to glycolytic HIF1α target genes, genes related to angiogenesis, also known as HIF1α target genes (Liao et al., 2007; Schödel et al., 2011), tend to be induced after exercise during the early active phase compared to the early rest phase (Figure 6B). Strikingly, HIF1α protein is significantly accumulated after exercise at the early active phase (Figure 6C). Thus, the time-of-day difference in the protein abundance of HIF1α after exercise also highlights distinct reliance upon glycolytic activation in skeletal muscle during exercise at different time of day.

We further determined the time-dependent impact of exercise on HIF1α activation by using bioinformatic tools and published high-throughput datasets. An interaction was reported between HIF1α and transcriptional co-activators CREB-binding protein (CBP)-p300, which is mainly thought to be required for the transcriptional activation by HIF1α (Dengler et al., 2014). Motif

Figure 5. Time-of-Day-Dependent Activation of Glycolytic Pathway after Exercise in Skeletal Muscle

(A) MS analysis of muscle glucose levels after exercise at the early rest phase (left) and active phase (right).

(B–D) Changes in blood glucose levels (B), muscle glycogen levels (C), and hepatic glycogen levels (D) after exercise at the early rest phase (left) and active phase (right).

Data are represented as mean ± SEM (n = 6) and were analyzed by two-way ANOVA using Bonferroni post hoc testing. *$p < 0.05$, **$p < 0.01$, and ***$p < 0.001$. The data were placed in order of time after exercise.

(E) Pathway analysis combining RNA-seq and metabolomics highlight acute impact of exercise at rest versus active phase on glycolytic pathway, TCA cycle, fatty acid oxidation, amino acid breakdown, and ketolysis in skeletal muscle. Changes in enzyme-encoding transcripts and metabolites in skeletal muscle from Exe mice compared to Sed mice at rest and active phases are indicated.

(F–H) Heatmaps displaying changes in metabolites and transcripts involved in glycolysis and TCA cycle (F), lipid oxidation (G), and BCAA catabolism (H) after exercise at rest versus active phase.

(I) Heatmaps displaying changes in transcripts involved in ketone body metabolism after exercise at rest versus active phase.

(J) MS analysis of β-hydroxybutylate (BHBA) levels in skeletal muscle after exercise at rest versus active phase. Data are represented as mean ± SEM (n = 6) and were analyzed by Student's t test. *$p < 0.05$.

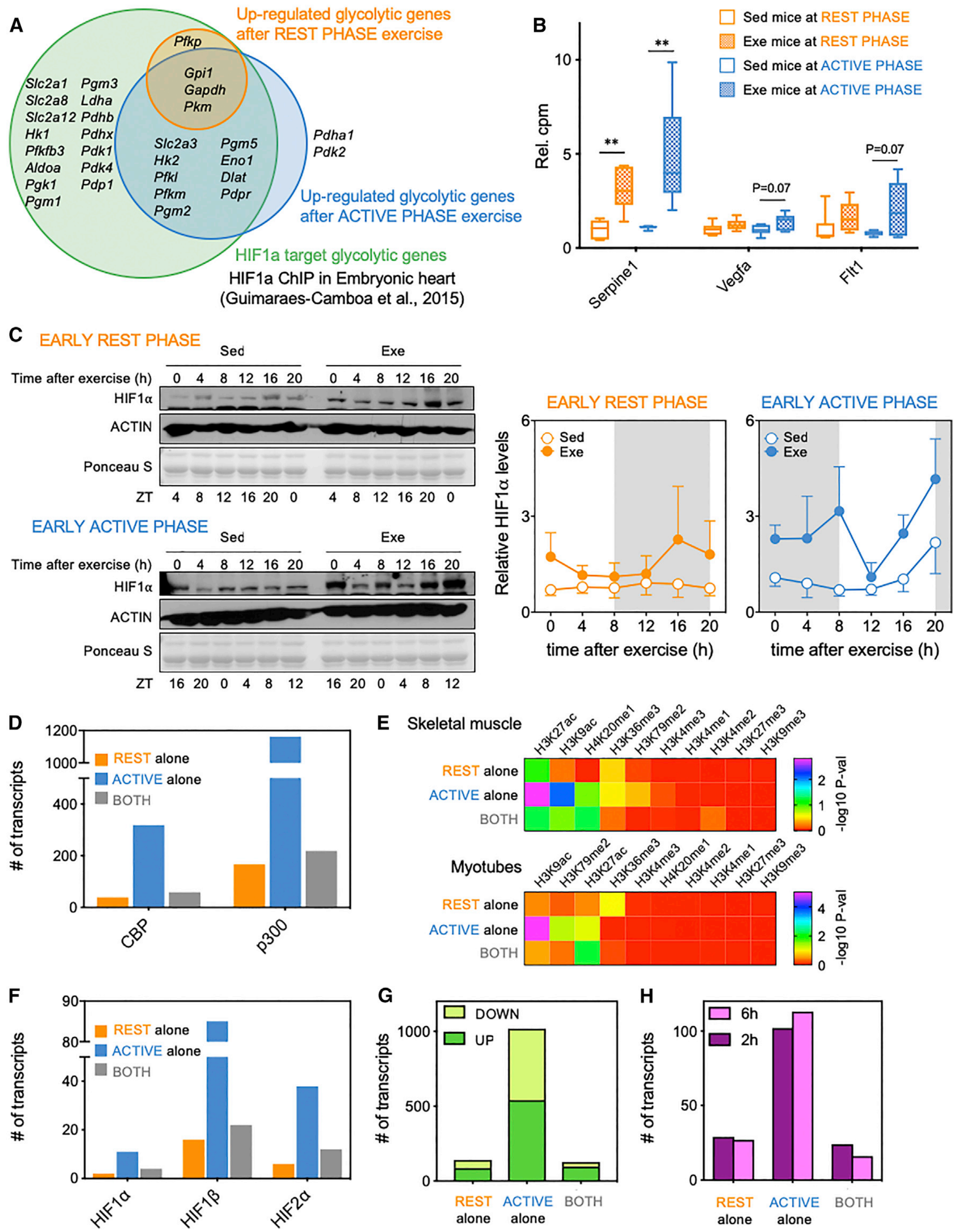

Figure 6. Activation of HIF1α in Skeletal Muscle after Exercise Preferentially at Early Active Phase

(A) Enrichment of HIF1α target genes within genes upregulated after exercise at the early rest phase versus active phase.

(B) The expression of HIF1α target genes involved in angiogenesis after exercise at the early rest phase (ZT3) versus early active phase (ZT15). Data are represented as mean ± SEM ($n = 6$) and were analyzed by Student's t test. **$p < 0.01$.

(C) Western blot images representing the daily protein abundance of HIF1α in skeletal muscle after exercise at rest (top) versus active phase (bottom). The levels of HIF1α were normalized to Ponceau S staining. Data are represented as mean ± SEM from four independent samples. The western blot data were placed in order of time after exercise.

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analysis revealed a higher enrichment of CBP-p300-bound target genes within genes upregulated after exercise at the active phase than after exercise at the rest phase (Figure 6D). As CBP-p300 have acetyltransferase activity, the activation of CBP-p300 leads to genome-wide histone acetylation (Dancy and Cole, 2015). In fact, comprehensive analysis of histone modifications (EnrichR [Chen et al., 2013; Kuleshov et al., 2016]) revealed higher enrichment of H3K9ac and H3K27ac target genes in skeletal muscle and myotubes within muscle transcripts exclusively induced by the active phase exercise (Figure 6E), suggesting a robust remodeling of epigenetic chromatin modifications after the early active phase exercise. Conclusively, motif analysis identified higher enrichment of transcripts targeted by the HIF family (HIF1α, HIF1β, and HIF2α) within muscle transcripts exclusively upregulated immediately after exercise at the active phase (Figure 6F). We compared a published HIF1α chromatin immunoprecipitation sequencing (ChIP-seq) dataset (Guimarães-Camboa et al., 2015) and a hypoxia-induced skeletal muscle transcriptome dataset (Gan et al., 2017) with the transcriptomic dataset from the present study (Figures 6G and 6H; Table S3). A 7.2-fold enrichment of HIF1α-bound genes is seen within the transcripts induced by exercise at the early active phase compared to the early rest phase (Figure 6G). Identically, transcripts induced by hypoxic exposure for 2 or 6 h are highly enriched in genes upregulated after exercise during the early active phase (Figure 6H). These observations indicate time-dependent activation of HIF1α pathway during exercise in skeletal muscle.

Distinct Impact of Exercise at Different Times of Day on Systemic Energy Homeostasis

Skeletal muscle regulates systemic energy homeostasis through humoral factors secreted not only from skeletal muscle but also from other metabolic tissues (Baskin et al., 2015). Of note, the activation of metabolic pathways in skeletal muscle during exercise improves whole-body energy metabolism, resulting in better metabolic health (Egan and Zierath, 2013; Vega et al., 2017). The accumulated evidence about the link between metabolic activation in skeletal muscle and systemic energy homeostasis prompted us to determine if the adaptation of systemic energy homeostasis to exercise is sensitive to time of exercise. We first measured the levels of humoral factors that drive skeletal muscle and systemic metabolic responses, including corticosterone in the muscle as well as serum cytokines and hormones. Although muscle corticosterone levels are drastically elevated immediately after the exercise bout at both early rest phase and early active phase (Figure S6A), there are virtually no time-dependent effects of exercise on the other factors (Figures S6A–S6K). Next, mice were subjected to sham exercise or exercise at ZT3 or ZT15 to measure the respiratory exchange ratio (RER), energy expenditure, food intake, and activity up to 60–72 h after exercise or sham exercise. RER and food intake are virtually unchanged following exercise at both phases (Figures 7A, 7B, S7A, and S7B). However, RER transiently drops immediately after exercise at the early active phase but not after exercise at the early rest phase (Figures 7A and 7B), suggesting the exhaustion of carbohydrate and hence the increased consumption of other energy sources. This correlates with our observations regarding the activation of glycolysis, lipid oxidation, amino acid breakdown, and ketolysis in skeletal muscle during the exercise at the early active phase. Similarly, energy expenditure of mice subjected to the early active phase exercise declines in parallel with the amount of activity soon after exercise (Figures 7D and S7D), suggesting extreme metabolic fatigue after exercise at this time point. Surprisingly, systemic energy expenditure is constantly higher in Exe mice than in Sed mice in the rest phase group, without remarkably altering food intake and ambulatory activity (Figures 7C, S7A, and S7C). Direct comparison of these parameters for systemic energy homeostasis uncovers a phase mismatch of rhythmic feeding activity followed by cyclic RER between Exe mice from the early rest phase versus the early active phase groups (Figures S7E–S7H). These findings demonstrate that the timing of exercise determines its impact on systemic energy homeostasis in addition to metabolic activation in skeletal muscle.

DISCUSSION

In the past decade, ‘‘time of food’’ has been recognized as a lifestyle modification to prevent metabolic disorders. In addition to what we eat and how much we eat, when we eat is a critical modifier of metabolic signaling pathways (Asher and Sassone-Corsi, 2015; Hatori et al., 2012). Increasing evidence suggests that cellular metabolism exhibits a robust daily oscillation. Indeed, several metabolic pathways are directly controlled by the circadian clock through genetic and epigenetic regulation of metabolic enzymes (Bass and Takahashi, 2010; Berger and Sassone-Corsi, 2016; Masri and Sassone-Corsi, 2014). Thus, daily timing of food intake, physical activity, sleep, and drug treatment could provide a diversity of metabolic and physiological responses and may influence disease pathogenesis. This concept has evoked interest in studies aimed to discover the optimal timing of physiological behaviors, as well as therapeutic treatment regimens to improve glucose and energy homeostasis.

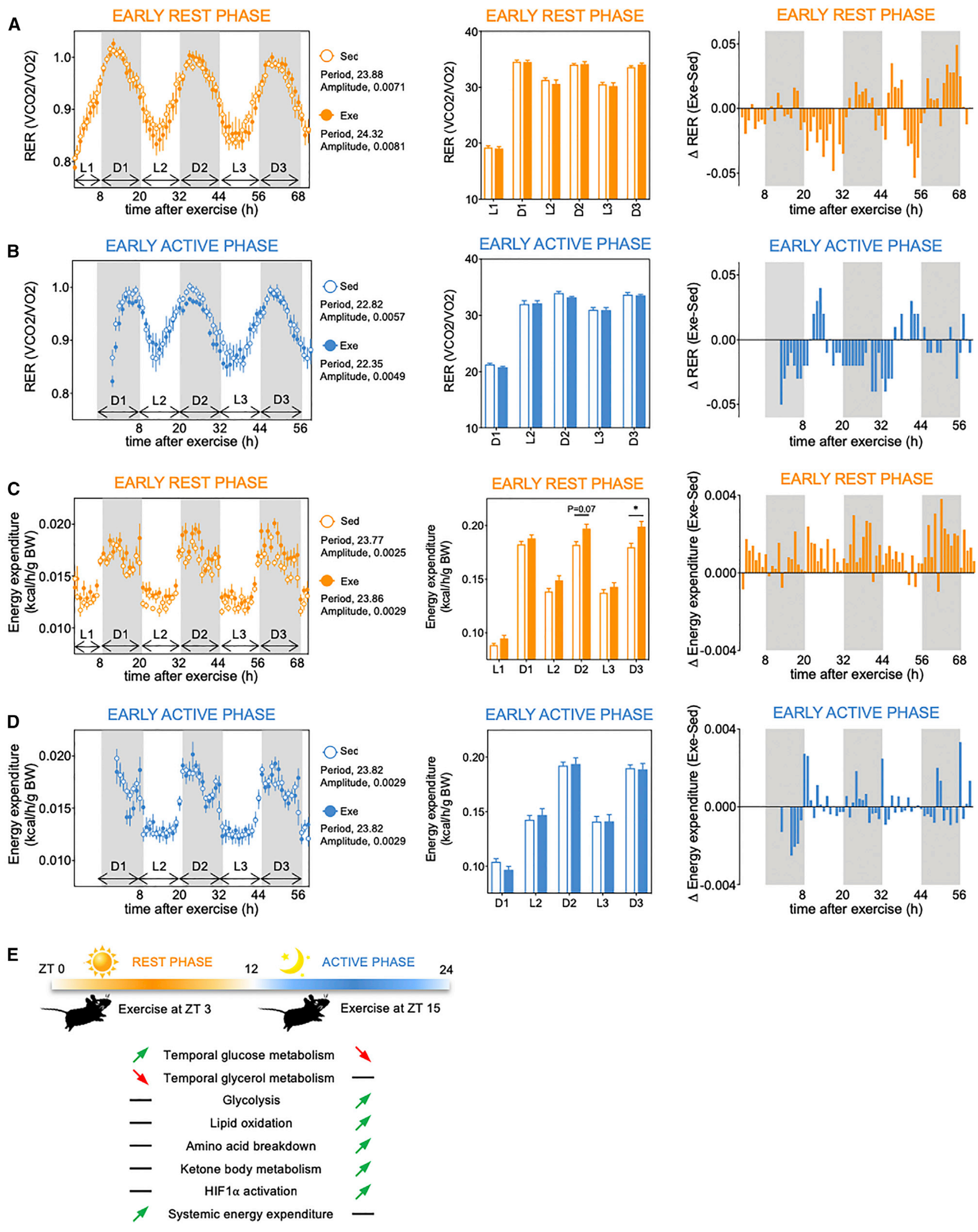
(D) MotifMap analysis comparing CBP- and p300-target genes versus genes upregulated by exercise at the early rest phase only (orange), active phase only (blue), and at both phases (gray).

(E) Comparative analysis of histone modifications in skeletal muscle (top) and myotube (bottom) enriched within transcripts exclusively induced by exercise at the early rest and active phases and transcripts commonly induced by exercise at both phases using EnrichR (http://amp.pharm.mssm.edu/Enrichr/) (Chen et al., 2013; Kuleshov et al., 2016).

(F) MotifMap analysis identifying the enrichment of target genes of HIF family within genes exclusively upregulated by exercise at rest phase (orange) and active phase (blue) and common upregulated genes (gray).

(G) The overlap between HIF1α-bound genes (Guimarães-Camboa et al., 2015) and downregulated or upregulated genes after exercise only at rest versus active phase and common to both phases.

(H) The overlap between genes induced by hypoxia for 2 or 6 h (Gan et al., 2017),genes exclusively induced after exercise at rest versus active phase, and commonly induced genes.



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Physical exercise provides multiple metabolic benefits, including increased insulin sensitivity, improved glucose metabolism, and body weight loss in humans (Camera et al., 2016; Fan and Evans, 2017). This not only emphasizes the importance of habitual exercise but establishes exercise as a non-pharmacological therapy for obese and diabetic patients. Therefore, efforts are ongoing to identify which types, intensity, duration, and frequency of exercise provide the most optimal metabolic benefits. In the present study, our findings underline that time of exercise could amplify the metabolic impact of exercise training (Figure 7E). Notably, Ezagouri et al. (2019) have determined the effect of time of exercise by using a different experimental paradigm, which complements our findings demonstrating the time-specific impact of exercise on metabolic pathways.

We have shown a time-of-day-dependent response of temporal metabolic pathways by combining temporal high-throughput transcriptomics and metabolomics. Exercise during the early rest phase (ZT3) stimulates the daily oscillation of genes and metabolites related to carbohydrate metabolism but dampens the daily rhythmicity of genes and metabolites related to glycerol metabolism. In contrast, rhythmic oscillation of genes and metabolites related to carbohydrate metabolism is lost after exercise during the early active phase (ZT15). In addition to the altered cyclic regulation of metabolic pathways in skeletal muscle, systemic oscillatory energy homeostasis including RER and energy expenditure exhibits distinct adaptations unique to the daily timing of exercise. These observations indicate that exercise modulates the molecular clock, as reported previously (Schroeder et al., 2012; Zambon et al., 2003), although its impact on the metabolic cycle appears to be dependent on the time of day. Our findings support the earlier evidence that food intake acts as a powerful *zeitgeber* to regulate the function of the metabolic clock (Asher and Sassone-Corsi, 2015; Vollmers et al., 2009). In addition, high-fat diet (HFD) and obesity reprograms the circadian control of metabolic pathways through cyclic activation of peroxisome proliferator-activated receptor γ (PPARγ) (Eckel-Mahan et al., 2013), PPARα, and sterol regulatory element-binding protein (SREBP) (Guan et al., 2018). In this study, we discovered a time-dependent impact of exercise on the activation of the hypoxia-responsible transcription factor HIF1α. Previous studies have demonstrated a role of HIF1α in regulation of the circadian clock (Adamovich et al., 2017; Peek et al., 2017; Wu et al., 2017). Together with our findings, this suggests that HIF1α acts as a key intermediator to induce timing-specific modification of temporal control of metabolic pathways by exercise.

Hypoxic conditions induce multiple adaptations in skeletal muscle, including angiogenesis, glycolytic activation, and suppression of mitochondrial function and biogenesis, through the activation of HIF1α (Favier et al., 2015; Formenti et al., 2010; Lindholm and Rundqvist, 2016). Our skeletal muscle transcriptomic analysis reveals that exercise in the early active phase dampens the levels of transcripts involved in mitochondrial respiratory functions but increases the level of transcripts related to angiogenesis and glycolysis. Targets of HIF1α, in conjunction with CBP-p300 and associated with histone hyperacetylation, are highly enriched genes exclusively upregulated by exercise at the early active phase compared to the early rest phase. Because HIF1α has been linked to molecular clocks (Adamovich et al., 2017; Peek et al., 2017; Wu et al., 2017), circadian variation of HIF1α activity could lead to distinct impact of exercise during the early active phase on its downstream signals. Our skeletal muscle metabolomic analysis reveals the depletion of carbohydrates only after the early active phase exercise, which is perfectly correlated with our transcriptomic findings of time-specific increase in glycolytic genes after the early active phase exercise. In addition to carbohydrates, skeletal muscle during endurance exercise utilizes various energy sources, including lipids, amino acids, and ketone bodies, upon the generation of ATP (Evans et al., 2017; Lundsgaard et al., 2018; Overmyer et al., 2015). The early active phase exercise robustly elevates the levels of muscle acyl-carnitines, BCAAs and BHBA, concomitantly with genes involved in fatty acid oxidation, BCAA catabolism, and ketolysis. Our findings clearly underscore the fact that energy utilization in response to exercise exhibits time-of-day specificity.

The loss of HIF1α in skeletal muscle leads to greater endurance capacity (Mason et al., 2004). Thus, HIF1α could be one of the limiting factors linking the metabolic energy production to endurance capacity during exercise. In addition to the molecular activation of HIF1α, skeletal muscle cells experience the most energy-depleted condition during the early active phase. Considering the daily variation of energy status associated with feeding-fasting cycle, exercise during the fasted period would deplete carbohydrate stores more quickly and shift energy utilization to lipid- and protein-based fuel in skeletal muscle. Of note, hepatic glycogen stores decline selectively after the early rest phase exercise, indicating tissue specificity of the time-dependent response of energy utilization to exercise. Changes in systemic energy homeostasis including RER and energy expenditure could be considered as the collective outcomes from metabolic adaptations in each tissue. This highlights the importance of studying the tissue-specific effect of different daily timing of exercise on metabolic reprogramming across the whole body. In conclusion, we propose that time of exercise dictates the amplitude of metabolic benefits from exercise, which potentially frames the basic strategy of chronobiology-based exercise therapy.

Figure 7. Effect of Different Timing of Exercise on Systemic Energy Homeostasis

(A and B) Circadian monitoring of RER of Sed and Exe mice at the early rest phase (ZT3; A) and at the early active phase (ZT15; B). Area under curve at each corresponding phase was calculated and displayed (middle). Values from Exe mice were subtracted from the values of Sed mice and are shown as delta RER (right).

(C and D) Circadian monitoring of energy expenditure of Sed and Exe mice at the early rest phase (ZT3; C) and at the early active phase (ZT15; D). Data represent energy expenditure normalized to body weight. Area under curve at each corresponding phase was calculated and displayed (middle). Values from Exe mice were subtracted from the values of Sed mice and are shown as delta energy expenditure (right).

Data are represented as mean ± SEM (n = 8) and were analyzed by two-way ANOVA using Bonferroni post hoc testing. *$p < 0.05$. The data were placed in order of time after exercise. Per and Amp indicate the period and amplitude of the oscillatory parameters, analyzed by Bio_Cycle, respectively.

(E) Schematic summary representing time-of-day-specific impact of exercise on skeletal muscle metabolism and systemic energy homeostasis.

Limitations of Study

Stringent circadian analyses involve sampling over the 24-h cycle with a frequency possibly shorter than every 4 h (Hughes et al., 2017; Li et al., 2015). The present study was designed to address the importance of time-of-day exercise, hence the limited sampling along the daily cycle. Investigating the effect of exercise at all times of the circadian cycle would require a highly complex and cumbersome experimental design that should include transcriptomic and metabolomic analyses also under constant dark conditions. Yet, our results clearly illustrate that the beneficial metabolic effect of exercise are time specific and hint at the possibility that exercise itself may operate as a *zeitgeber* for the muscle circadian system.

Interestingly, while the impact of acute exercise on the HIF1α pathway in the skeletal muscle has been reported (Lindholm and Rundqvist, 2016), a central finding of the present study is that HIF1α activation is dependent on daily timing of exercise (Figure 6C) as HIF1α is selectively activated by exercise at the early active phase. On the other hand, we found that energy expenditure significantly rises following a single bout of exercise exclusively at the early rest phase (Figure 7C), Finally, time of exercise may affect various organs in different manners, as indicated by the glycogen levels in the liver in response to exercise (Figure 5D), unparalleled with a drastic decrease of muscle glycogen levels after the early active phase exercise (Figure 5C). Further studies will be required to reveal metabolic communications across tissues and time under exercise, leading to a better understanding of the beneficial effects of exercise.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental Information can be found online at https://doi.org/10.1016/j.cmet.2019.03.013.

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AUTHOR CONTRIBUTIONS

S.S., A.L.B., M.S., J.T.T., J.R.Z., and P.S.-C. designed all experiments. S.S., A.L.B., M.S., R.C.L., and E.D. performed experiments. S.S., A.L.B., S.C., M.S., A.A., R.B., P.B., and J.T.T. analyzed data. S.S., J.R.Z., and P.S.-C. prepared the figures and wrote the manuscript. All authors read and approved the final version of the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti-HIF1α	Abcam	Cat# ab2185; RRID: AB_302883
Anti-ACTIN	Abcam	Cat# ab3280; RRID: AB_303668
Anti-Mouse IgG, HRP conjugate	EMD Millipore	Cat# AP160P; RRID: AB_92531
Anti-Rabbit IgG, HRP-linked	Cell Signaling Technology	Cat# 7074; RRID: AB_2099233
Chemicals, Peptides, and Recombinant Proteins		
Hexokinase/Glucose-6-Phosphate Dehydrogenase	Roche Diagnostics	Cat# 10737275001
Protein Assay Dye Reagent	Bio-Rad Laboratories	Cat# 500-0006
Nitrocellulose Membrane, 0.45 μm	Bio-Rad Laboratories	Cat# 1620115
Immobilon Western Chemiluminescent HRP substrate	EMD Millipore	Cat# WBKLS0500
cOmplete, EDT-free Protease Inhibitor Cocktail	Roche	Cat# 11873580001
Critical Commercial Assays		
AllPrep DNA/RNA/miRNA Universal Kit	Qiagen	Cat# 80224
TruSeq Stranded Total RNA LT Sample Prep Kit with Ribo-Zero Gold	Illumina	Cat# 20020596
Agilent RNA 600 nano kit and Bioanalyser instrument	Agilent Technologies	Cat# 5067-1511
Qubit dsDNA HS assay kit	Invitrogen	Cat# Q32851
Agilent High Sensitivity DNA chip	Agilent Technologies	Cat# G2938-90321
Glucometer	Bayer	Contour Next EZ
V-Plex Proinflammatory Panel 1 (mouse) Kit	Meso Scale Discovery	Cat# K15048G-1
Mouse Metabolic Kit	Meso Scale Discovery	Cat# K15124C-1
Deposited Data		
Transcriptomics data	This paper	GEO# in preparation
ChIP-seq data	(Guimarães-Camboa et al., 2015)	GEO: GSE61247
Transcriptomics data	(Gan et al., 2017)	GEO: GSE81286
Experimental Models: Organisms/Strains		
C57B6/J mice	Taconic Biosciences	N/A
Software and Algorithms		
Prism 7.0	GraphPad Software	http://www.graphpad.com
FastQC (v0.11.5)	Babraham Institute	http://www.bioinformatics.babraham.ac.uk/projects/fastqc/
Trim_Galore (v0.4.3 using Cutadapt v1.16)	Babraham Institute	https://www.bioinformatics.babraham.ac.uk/projects/trim_galore/
STAR (v2.5.3a)	(Dobin et al., 2013)	https://github.com/alexdobin/STAR
featureCounts (v1.5.2)	(Liao et al., 2014)	http://subread.sourceforge.net/
edgeR (v3.2.23)	(Robinson et al., 2010)	https://bioconductor.org/packages/release/bioc/html/edgeR.html
Bio_Cycle	(Agostinelli et al., 2016)	http://circadiomics.igb.uci.edu/biocycle
ImageJ	NIH	https://imagej.nih.gov/ij/
EnrichR	(Chen et al., 2013; Kuleshov et al., 2016)	http://amp.pharm.mssm.edu/Enrichr/

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
CircadiOmics	(Ceglia et al., 2018; Patel et al., 2012)	circadiomics.igb.uci.edu
MotifMap	(Daily et al., 2011; Xie et al., 2009)	http://motifmap.ics.uci.edu/
Bioconductor	(Gentleman et al., 2004)	https://www.bioconductor.org/
Other		
Animal treadmill	Columbus Instruments	Exer 3/6
Home Cage System for indirect calorimetry	TSE Systems	PhenoMaster
Activity monitor	TSE Systems	ActiMot

CONTACT FOR REAGENT AND RESOURCE SHARING

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Paolo Sassone-Corsi (psc@uci.edu).

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Specific Pathogen Free (SPF) male C57B6/JBomTac mice were purchased from Taconic Biosciences and maintained at the animal facilities of the University of Copenhagen, Denmark. Animal experiments complied with the European directive 2010/63/EU of the European Parliament and were approved by the Danish Animal Experiments Inspectorate (2012-15-2934-26 and 2015-15-0201-796). Mice were maintained on a 12 hr light/dark cycle.

METHOD DETAILS

Animal Experiments

Mice were reared at 22±1°C with 12 hr light/dark cycle (light on at 6AM (ZT0) and light off at 6PM (ZT12)) and fed a standard rodent laboratory diet (#1310, Altromin, Germany) ad libitum. At 10-11 weeks of age, the mice were divided into two different groups of exercise or sham-exercise treatment either during the early rest phase group (ZT3) or the early active phase group (ZT15). Exercised mice in the early rest phase group were subjected to a single-bout of acute exercise for 1 hour at ZT3, whereas exercised mice in the early active phase group were subjected to the exercise for 1 hour at ZT15. Sedentary control mice for corresponding exercised mice at the different daily times were placed on a stationary treadmill for 1 hour (sham-exercise). Mice were returned into their home cages and allowed free feeding and drinking after the exercise and sham-exercise treatment. At 0, 4, 8, 12,16 and 20 hours following exercise or sham-exercise treatment at each early rest and active phase, mice were sacrificed under isoflurane anesthesia and the gastrocnemius and quadriceps muscles, and livers were isolated. The tissues were immediately snap frozen in liquid nitrogen and stored at -80°C for subsequent use.

Exercise Protocol

The mice were subjected to a single-bout of acute treadmill running (Columbus Instruments, OH) following 4-day's acclimatization protocol as described below (Brandauer et al., 2015; Treebak et al., 2014);

Day 1 (15 min Total; 5% Incline)

For the first 5 min, mice were placed on the stationary treadmill.Start running at 6 m/min and accelerate by 2 m/min every 2 or 3 min up to 12 m/min

Day 2 (15 min Total; 5% Incline)

Start running at 6 m/min and accelerate by 2 m/min every 3 min up to 14 m/min

Day 3 (15 min Total; 5% Incline)

Start running at 6 m/min and accelerate by 2 m/min every 2 or 3 min up to 16 m/min

Day 4

Rest

Day 5 (60 min Total; 5% Incline)

Start running at 6 m/min and accelerate by 2 m/min every 2 min up to 16 m/min. Electrical grids forced mice to keep running for the corresponding duration of exercise.

RNA Sequencing

Gastrocnemius muscles from both legs of each mouse were ground up in liquid nitrogen to carry out RNA-seq analysis (total 144 individual muscle samples, n=6 per time point). Sample-size estimation was not done. RNA was extracted using AllPrep DNA/RNA/miRNA Universal Kit and was checked for quality using the Bioanalyzer instrument (Agilent Technologies, Santa Clara, CA) and subject to the Illumina TruSeq Stranded Total RNA with Ribo-Zero Gold protocol (Illumina, San Diego, CA) and performed as described. Libraries were subjected to 100-bp paired-end sequencing on HiSeq 2500 (Illumina) at the Beijing Genomics Institute (BGI, Hong Kong). On average, 12 million reads/sample were assigned to a genomic annotation with 11,461 genes surviving the expression threshold. The depth of sequencing was considered based on the study demonstrating that the biological replication is more important than the sequencing depth for differential expression analysis (Liu et al., 2014). RNA-seq reads were subjected to trimming of adapters and low quality flanking ends using Trim Galore and Cutadapt. Pre-processed reads were mapped to mm10 using STAR aligner (Dobin et al., 2013) and gene coverages were computed with featureCounts (Liao et al., 2014) and the Gencode annotation (GRCm38). The list of RNAs was filtered for those with a read coverage larger than 2 CPM (counts per million) in at least 34 samples, and 11,461 of the total 52,636 annotated RNAs survived this inclusion criterion. The edge R pipeline for differential expression analysis and the CPM calculation considered different library sizes and runs a normalization by using TMM (trimmed mean of M-values algorithm). The results of statistics of the RNA sequencing analysis are available through the supplemental HTML file (Data S4). At the end, 4 samples were excluded from the analysis since there was a corruption in the fastq files.

Gene Ontology Analysis

GO analysis of the temporal transcriptome was performed using Pathway Commons (http://www.pathwaycommons.org), a web resource for biological pathway data (Cerami et al., 2011). GO analysis of the enzymes linked to metabolites was carried out using Gene Ontology Consortium (http://geneontology.org/).

Motif Analysis

MotifMap, a comprehensive database of putative regulatory transcription factor binding sites (Daily et al., 2011; Xie et al., 2009), was applied to the motif analysis as described previously (Eckel-Mahan et al., 2012, 2013). Briefly, each position weight matrix is used to scan the mouse genome for potential binding sites with every site being assigned the following three scores: (1) motif matching score (Z-score); (2) conservation score (Bayesian Branch Length Score or BBLS); and (3) false discovery rate (FDR).

Mass Spectrometry for Metabolomics Analysis

Metabolomic analysis was performed by Metabolon, Inc. (Durham, NC) as previously described (Evans et al., 2009). Small biochemicals from 6 independent skeletal muscle (per ZT and experimental group) were methanol extracted and analyzed by ultra-high-performance liquid chromatography-tandem mass spectrometry (UPLC-MS/MS; positive mode), UPLC-MS/MS (negative mode) and gas chromatography-mass-spectrometry (GC-MS). Metabolites were identified by automated comparison of the ion features in the experimental samples to a reference library of chemical standard entries that included retention time, molecular weight (m/z), preferred adducts, and in-source fragments as well as associated MS spectra and were curated by visual inspection for quality control using software developed at Metabolon, Inc. (Dehaven et al., 2010).

Correlation Analysis between Temporal Skeletal Muscle Transcriptome and Metabolome

To perform correlation analysis between the temporal skeletal muscle transcriptome and metabolome, the datasets containing expression levels for the transcriptome and metabolome were run through Bio_Cycle (Agostinelli et al., 2016) via the CircadiOmics web portal (Ceglia et al., 2018; Patel et al., 2012). The output is then filtered using a p-value of 0.05 to include only enzymes and metabolites that are significantly periodic. A pairwise analysis of these enzymes and metabolites is then performed to determine interactions. For each enzyme-metabolite pair, where each element in the pair is oscillating, a score indicative of a potential interaction is calculated. This score is calculated via the CircadiOmics web portal and incorporates data from databases including KEGG (Kanehisa et al., 2016, 2017; Kanehisa and Goto, 2000), enzyme-metabolite occurrences in literature, and evidence from mass spectrometry experiments. The higher the interaction score, the more likely there is an interaction between the enzyme and metabolite. The output of this interaction analysis is then filtered to only include scores from the 50th percentile and above. The pairs of enzymes and metabolites associated with these high scores are then analyzed to determine what pathways they share. The list of metabolic enzymes in Sed and Exe mice from each phase group was analyzed with Gene Ontology Consortium (http://geneontology.org/) for functional enrichment pathway analysis.

Measurement of Blood Glucose Levels

At 0, 4, 8, 12, 16 and 20 hours following exercise, the mice were subjected to blood collection from the tail vein before anesthesia and glucose concentrations were measured by using glucometers.

Measurement of Hepatic Glycogen and Muscle Glycogen Levels

Glycogen was extracted from mouse liver and quadriceps muscle samples by acidic extraction using 1M HCl at 95°C for 2h. After a quick spin tissue extracts were neutralized using 2 M NaOH followed by centrifugation at 16,000 g for 20 min at 4°C . The supernatant was transferred to new tubes and a reaction mix consisting of 200 mM Tris-HCL, 500 mM $MgCl_2$, 5.2 mM ATP, 2.8 mM NADP and 6 μg/ml of a hexokinase and glucose-6-phosphate dehydrogenase mixture (#10737275001, Roche Diagnostics, Mannheim, Germany) were added. After 15 min of incubation at RT, the glucose generated from the hydrolyzed glycogen were measured spectrophotometrically at 340 nm (Hidex sense, Hidex, Finland). All samples were measured in triplicates along with glucose standards.

Western Blot

Whole cell lysates were extracted from skeletal muscle using the buffer containing 137mM NaCl, 2.7mM KCl, 1mM MgCl2, 5mM Na4P2O7, 10mM NaF, 1% Triton X-100, 10% glycerol, 20mM Tris pH7.8, 1mM EDTA, 0.2mM PMSF, 0.5mM Na3VO4, 10mM NAM, 0.33uM TSA, and 1X protein inhibitor cocktail (Roche). Forty ug of lysates was applied to SDS-PAGE. Anti-HIF1α (Abcam, ab2185) and anti-ACTIN (Abcam, ab3280) antibodies were used for western blot.

Enrichment Analysis of Transcriptome Analysis

The transcriptome dataset in this study was crossed with published HIF1α ChIP-seq dataset (GSE61247) (Guimarães-Camboa et al., 2015) that includes HIF1α bound genes and transcriptome dataset (GSE81286) (Gan et al., 2017) that includes genes up-regulated or down-regulated under hypoxia. EnrichR (http://amp.pharm.mssm.edu/Enrichr/), a comprehensive gene set enrichment analysis web tool (Chen et al., 2013; Kuleshov et al., 2016), was applied to a comparative enrichment analysis between the skeletal muscle transcriptome in this study and ChIP-seq datasets in skeletal muscle or myotube for histone H3K9ac, H3K27ac, H3K4me1, H3K4me2, H3K4me3, H3K9me3, H3K27me3, H3K36me3, H3K79me2 and H4K20me1.

Measurement of Serum Hormones and Cytokines

Using frozen serum samples, IFN-γ, IL-1β, IL-2, IL-5, IL-6, IL-10, KC/GRO and TNF-α were measured by the V-Plex Proinflammatory Panel 1 (mouse) Kit from Meso Scale Discovery (Rockville, MD), according to the manufacture's protocol. Leptin and insulin were quantified in the serum using the Mouse Metabolic Kit from Meso Scale Discovery, according to the manufacture's protocol.

Indirect Calorimetry

Indirect calorimetry was performed using Home Cage System Phenomaster (TSE Systems, Bad Homburg, Germany). Briefly, animals were acclimated in training cages for 4 days prior to the measurement and were allowed 4 additional days to acclimate to the TSE cabinets. During this period the mice were also acclimated to the treadmills using a similar exercise protocol as described above. However, to obtain baseline measurements the rest period was extended from 1 to 3 days. After a 1 hour bout of acute exercise or sham-exercise at ZT3 or ZT15, mice were returned to their home cages and gas exchanges and food intake were recorded every 20 minutes over 72 hours. Activity monitoring was conducted using an infrared light beam activity monitor (ActiMot, TSE Systems, Bad Homburg, Germany).

QUANTIFICATION AND STATISTICAL ANALYSIS

Quantification

Western blots for HIF1α and ACTIN antibody were quantified using ImageJ (NIH).

Statistical Analysis

Data are represented as mean±SEM and were analyzed by Student's t-test or Two-way ANOVA with Bonferroni post hoc testing where applicable (Prism 7.0). For the analysis of rhythmic genes and metabolites, the nonparametric test Bio_Cycle was used incorporating a window of 20-28 hr for the determination of circadian periodicity (Agostinelli et al., 2016), including amplitude and phase analysis. A gene or metabolites was considered circadian based on a p-value cutoff of 0.05.

DATA AND SOFTWARE AVAILABILITY

RNA sequencing data reported in this paper has been deposited into NCBI's Gene Expression Omnibus under accession number GEO: GSE126962.